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Attenuation of *Acinetobacter baumannii* virulence by inhibition of polyphosphate kinase 1 with repurposed drugs

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Abstract

Acinetobacter baumannii is clinically one of the most significant pathogens, especially in intensive care settings, because of its multidrug-resistance (MDR). Repositioning of high-affinity drugs is a faster and more plausible approach for combating the emergence of MDR and to tackle bacterial infections. This study was aimed to evaluate the approved drugs potentially inhibiting *A. baumannii* PPK1 (AbPPK1) mediated synthesis of polyphosphates (polyP). Based on virtual screening, molecular dynamic simulation, and CD spectroscopy for thermal stability, two stable ligands, etoposide and genistein, were found with promising contours for further investigation. Following *in vitro* inhibition of AbPPK1, the efficacy of selected drugs was further tested against virulence traits of *A. baumannii*. These drugs significantly reduced the biofilm formation, surface motility in *A. baumannii* and led to decreased survival under desiccation. In addition to inhibition of PPK1, both drugs increased the expression of polyP degrading enzyme, exopolyphosphatase (PPX), that might be responsible for the decrease in the total cellular polyP. Since polyP modulates the virulence factors in bacteria, destabilization of the polyP pool by these drugs seems particularly striking for their therapeutic applications against *A. baumannii*.

Keywords: Repurposed drugs, virulence, biofilm, PPK1 inhibition, genistein, etoposide, desiccation, MD Simulation.

1. Introduction

Acinetobacter baumannii has emerged as an opportunistic Gram-negative, multidrug-resistant pathogen prevalent in hospital settings. The ability to resist desiccation, antibiotics, and other environmental stresses makes it a resilient organism (Doi et al., 2015; Rajamohan et al., 2010). Metabolic adaptability of *A. baumannii* provides it a selective advantage in hospital settings (Peleg et al., 2012). It can form biofilm on abiotic and biotic surfaces and can persist in ICU, thus posing a challenge for the treatment of immunocompromised patients (Gaddy et al., 2009). *A. baumannii* is resistant to most of the antimicrobial drugs. Recently, the term "extensively drug-resistant (XDR)" has been used for characterization of *A. baumannii* isolates, which are resistant to all antibiotics except tigecycline and polymyxins (Asif et al., 2018). The ability of *A. baumannii* strains to continuously acquire resistance to newly developed antimicrobial drugs has catapulted it to the top of the list of priority pathogens for which the development of new antibiotics is required (WHO, 2017). Tolerance to desiccation is one of the important features of the nosocomial pathogens (Farrow et al., 2018), and *A. baumannii* is reported to survive up to five months on dry surfaces, posing a threat to hospital infection control measures (Kramer et al., 2006).

Polyphosphate kinase 1 (PPK1) (EC 2.7.4.1) is the principal enzyme that catalyzes the reversible synthesis of polyphosphates from ATP. It helps microorganisms to cope up with stringent environment and is responsible for the virulence and survival of pathogens (Candon et al., 2007; Rudat et al., 2018; Wang et al., 2018). In major human pathogens, the deletion of the *ppk* gene has revealed its essential role in pathogenesis, quorum sensing, motility, and biofilm formation, which contributes largely to antibiotic resistance (Chen et al., 2016; Rashid and Kornberg, 2000). Furthermore, absence of *ppk*1 orthologue in humans makes it potential for the design and development of new therapeutics (Gautam et al., 2019; Zhang et al., 2007). The crystal structure of *E. coli* PPK and its complex with β-γ-imido-adenosine 5-phosphate (AMP-PNP) provided an insight into the mode of binding of ATP in the PPK dimer tunnel and the mechanism of polyP elongation (Zhu et al., 2003, 2005). Like most pathogenic bacteria (Gangaiah et al., 2009; Ishige et al., 1998; Zhu et al., 2005), *A.*

baumannii also encodes PPK1 homolog in its genome, which needs to be characterized and explored as a possible drug target.

In the present study, *Acinetobacter baumannii* PPK1 (AbPPK1) model structure was generated by homology modeling based on *E. coli* PPK1 and was used for the screening of potential inhibitors from chemical libraries. Among the screened compounds, etoposide, which is being used for the treatment of multiple myeloma, and genistein, a phytoestrogen with therapeutic properties, effectively reduced the PPK activity *in vitro* and were found to decreased the polyP accumulation, biofilm formation, motility, and survival under desiccation in *A. baumannii*. To the best of our knowledge, this is the first study where approved drugs have been shown to inhibit the *Ab*PPK1 activity and attenuate virulence factors, warranting *in vivo* validation to repurpose their antibacterial effects against *A. baumannii*.

2. Materials and Methods

2.1 Sequence analysis and evolutionary relatedness

The amino acid sequence of *Ab*PPK1 obtained from the Uniprot database (https://www.uniprot.org/) was subjected to protein BLAST (BLASTp) to identify the protein family. This program used the BLOSUM 62 scoring matrix for similarity and identity search. Clustal Omega (https://www.ebi.ac.uk/Tools/msa/clustalo/) and the Multalin program (http://multalin.toulouse.inra.fr/multalin/) were used for sequence alignment and analysis. Evolutionary divergence of PPK1 was analyzed among pathogens using MUSCLE (https://www.ebi.ac.uk/Tools/msa/muscle/), and the phylogenetic tree was generated using PhyML (Guindon et al., 2010) with bootstrap value of 100, using FigTree on phylogeny.fr server (Dereeper et al., 2008).

2.2. Homology modeling and active site prediction

The homology model of AbPPK1 was prepared based on *E. coli* PPK1 (PDB: 1XDP) template structure using the I-Tasser webserver (Roy et al., 2010). The 3D model was evaluated for biophysical and stereo-chemical properties by Verify-3D, ERRAT, and PROCHECK at SAVES server (http://servicesn.mbi.ucla.edu/SAVES/). Bond orders and missing atoms were corrected in the model, and the system was energy minimized. Binding site prediction was performed using 3Dligandsite (Wass et al., 2010), SAS (Milburn et al., 1998), and I-Tasser servers (Roy et al., 2010). ATP was docked in the active site of *Ab*PPK1 by using the coordinate information to generate the *Ab*PPK1-ATP complex. All the protein-ligand complexes were inspected for structure artifacts.

2.3. Virtual screening and molecular docking

684 natural compounds and 453 FDA approved drugs chosen based on data mining were retrieved from PubChem (Bolton et al., 2008) and ZINC (http://zinc.docking.org) databases for screening against AbPPK1. OpenBabel Graphical User Interface (GUI) and PyRx were used to convert the 3D structures of ligands to different formats (Dallakyan and Olson, 2015; O'Boyle et al., 2011). Kollman and Gasteiger charges were assigned to the ligands. AbPPK1 and ligands were energy minimized and exported to Autodock compatible format. Autodock Tools v1.5 was used to set up the docking environment using its default parameters (Trott and Olson, 2009). Rigid docking parameter was applied for AbPPK1, and ten poses of ligands were allowed. The grid box size was defined to 30 × 30 × 22 point size, in X, Y, and Z direction, respectively, with 0.375Å spacing to accommodate and dock ligands perfectly into the active site. The docking exhaustiveness limit was set to 10. After inspecting top scoring molecules, compounds were prioritized on the basis of their contact with key catalytic residues. H-bonding, hydrophobic interactions, and solvent accessibility for all structure complexes were evaluated using PyMol viewer, Discovery Studio (Dassault Systèmes) and Maestro 10.5 suite (Schrödinger LLP).

2.4. Molecular Dynamics Simulation

The model structures of *Ab*PPK1 complexed with ATP and selected ligands were simulated for their stability. MD simulation was performed for 20 ns with GROMACS 5.1.2 by employing the Amber force field (Van Der Spoel et al., 2005). The ligand topology file with an appropriate force field was generated from the PRODRG server (Schüttelkopf and Van Aalten, 2004). Structures were appended in a rhombic dodecahedron box filled with the TIP3P water model. The distance between the protein and the inner wall of the box was set to 1.0. Neutralization of the charge of the system was done by adding ions (Na^+ / Cl^-) to the complex. The energy of the system was relaxed with the steepest descent energy minimization. The simulation was performed using LINear Constraint Solver (LINCS) algorithm to constrain bond length. Periodic boundary condition (PBC) was applied in all directions. All simulations were equilibrated in NVT and NPT environment for temperature coupling with reference temperature set at 300K. VMD (Version 1.9.3**)** was used for viewing, editing, and analyzing the trajectory, and gedit was used to edit files and scripts.

2.5. Organisms and culture conditions

A. baumannii ATCC 17978, *A. baumannii* clinical isolates, *E. coli* DH5α, *E. coli* BL21 (DE3), and *Agrobacterium tumefaciens* A136 were grown in Lysogeny Broth (LB) at 37°C under shaking conditions or indicated otherwise.

2.6. Cloning of the ppk1 gene

*ppk*1 gene sequence of *A. baumannii* ATCC 17978 was retrieved from NCBI (https://www.ncbi.nlm.nih.gov). *ppk*1 gene was amplified from genomic DNA using forward 5’AAAGGATCCATGAATACAGCGATTACA and reverse 5’AATGCGGCCGCTTATTTAAAAGTTTCTAATAA primers. Amplified product and pET28a were digested with BamHI and NotI restriction endonucleases, ligated, and transformed into the *E. coli* DH5α electro-competent cells. The recombinant pET28a-*ppk*1 was isolated, and the insert was confirmed by sequencing. The recombinant plasmid was further transformed into *E. coli* BL21 for expression and purification of AbPPK1.

2.7. Generation of pET28a-ppk1^{H443P} and pET28a-ppk1^{H601A}

Q5 Quick-Change mutagenesis kit (NEB, England) was used for substitution mutation in the recombinant plasmid (pET28a-*ppk*1). Primer pairs were custom-designed, Fw-5’-GTTATAAAACCCCTGCCAAAATGATT/Rv-5’-AATCATTTTGGCAGGGGTTTTATAAC and Fw-5’-GTTTCCTTCGAAGCTACTCGAGTTTA/Rv-5’-TAAACTCGAGTAGCTTCGAAGGAAAC to change histidine (CAT) to proline (CCT) and alanine (GCT) at positions 1329 and 1803 respectively in the *ppk*1 sequence in pET28a-*ppk*1 to generate pET28a-*ppk*1^{H443P} and pET28a-*ppk*1$^{H443P-H601A}$.

2.8. Expression and purification of AbPPK1

E. coli BL 21 (DE3) cells transformed with pET28a-*ppk1*, pET28a-*ppk*1^{H443P}, and pET28a-*ppk*1$^{H443P-H601A}$ were induced with 0.1 mM IPTG for 5 h at 28 ° C and 180 rpm. Cells were harvested, resuspended in the buffer (50mM Tris-HCl, 150mM NaCl), and disrupted by sonication (25% amplitude and 10s on/off cycle for 30 min). The lysate was processed and AbPPK1 was purified from the supernatant using Ni^{2+}-NTA agarose column (Qiagen, Germany) and checked on 10% SDS PAGE. Purified protein was dialyzed, quantitated using the Bradford assay kit (Sigma) taking BSA as the standard. Mutant AbPPK1 preparation was also processed in the same way.

2.9. CD spectroscopy for in vitro stability of AbPPK1

Isothermal wavelength scan of AbPPK1 and its mutants was carried out with JASCO-J-815 spectro-polarimeter connected to a Peltier-temperature controller. The path length of the cell was 0.1 cm for Far- UV CD (200 - 250 nm). A single scan baseline was obtained with buffer (20 mM Tris-Cl, 50 mM NaCl, pH 7.2). The relative content of α-helices and β-sheets was calculated by processing the Far UV CD spectra using the BestSel online tool. Thermal denaturation of AbPPK1, with and without the ligands was performed by recording the changes in ellipticity at 222 nm as a function of temperature (1°C per minute) at 30 to 80°C. ATP, etoposide, and genistein were used at a molar concentration of 100 µM with a protein: ligand ratio of 1:20, and the reaction mix was incubated for 30 min before performing a thermal denaturation scan.

2.10. Phosphotransferase activity

Specific activity (µM min^{-1} mg^{-1}) of AbPPK1 was determined using an ADP quantification kit (AAT Bioquest, Inc.). Conversion of ATP to ADP by *Ab*PPK1 was carried out in reaction buffer comprising 50 mM Tris-HCl (pH 7.2), 150 mM NaCl, 1 mM ATP, 1 mM $MgCl_2$, and 1 µM *Ab*PPK1 incubated for 30 min at 25°C (Lander et al., 2013). Immediately after incubation, components A and B from the kit were added and incubated at 25°C for 30 min. Fluorescence was detected at $540^{ex}/590^{em}$ nm by BioTek Synergy H1 multimode plate reader. ADP standard curve (0.05 - 10µM) was prepared for quantification purposes. GraphPad Prism was used for data analysis and plotting the graph.

2.11. Real-time PCR

Change in expression of *ppk*1, *ppx, pho*U and *rpo*S in the presence of drugs was evaluated using gene-specific primers (Table S1). Late log phase culture was treated with drugs (200µM in DMSO) for 1 h and RNA was extracted b TRIzol (Rio et al., 2010). cDNA was synthesized from RNA using the Verso cDNA kit (Thermo Scientific). 16S ribosomal RNA gene was used as the internal control. qRT PCR was conducted using the CFX 96 Real-Time PCR (Bio-Rad). For each gene, a minimum of three biological and two technical replicates was performed, and the relative fold expression was calculated by the $2^{-\Delta\Delta ct}$ method (Pfaffl, 2004). The data were analysed by one way ANOVA, using treatments as factors. Tukey's test was used to detect significant differences ($p \leq 0.05$) among treatments in pairwise comparison.

2.12. Acyl homoserine lactone (AHL) production

Acyl homoserine lactone (AHL) was extracted from the cell-free supernatant (CFS) of *A. baumannii* culture was grown in low salt LB broth under static conditions in the presence and absence of drugs. The CFS was mixed with an equal volume of acidified (acetic acid, 0.01% v/v) ethyl acetate and incubated overnight at 4˚C. The aqueous phase containing AHL was separated and dried using a rotary evaporator and re-dissolved in dimethyl sulphoxide (DMSO) (Shaw et al., 1997). *A. tumefaciens* A136 biosensor was grown for 16 h at 30˚C, and β-galactosidase activity in the presence of extracted AHL was calculated (Zhu et al., 2003). The standard curve was prepared with N-(3-Oxododecanoyl)-L-homoserine lactone (10 - 100µM) for the quantitation of AHL.

2.13. Virulence factors

PolyP: A. baumannii culture was grown for 3, 5, and 24 h in the presence and absence of drugs for quantitation of polyP (Grillo-Puertas et al., 2012). Cells were harvested, washed, and resuspended in 100 mM Tris HCl, pH 7.5, followed by the addition of chloroform and 0.1 % SDS. 20 µM DAPI was added to the suspension and agitated for 5 min at 37°C, and fluorescence spectra ($415^{ex}/550^{em}$ nm) were recorded using Synergy H1 multimode plate reader.

Biofilm formation: Drugs at different concentrations (50, 100, and 200µM) were evaluated for its effect on biofilm formation by inoculating 5% overnight grown *A. baumannii* culture

(10^8 CFU/ml) in 10 ml LB with glycerol (1% v/v) in polystyrene tubes under the static condition at 37°C for 24 h. Change in the biofilm index (BI) was determined as OD^{570nm} /OD^{600nm} (Babić et al., 2010).

Determination of MICs

The MIC of different classes of antibiotics (rifampicin, tobramycin, ciprofloxacin, and colistin) against *A. baumannii* was determined by the microdilution method as per CLSI guidelines, 2017. The concentration at which no visible growth was observed as taken as MIC. The effect of different concentrations of antibiotics (3×, 5×, and 10× MIC) alone and in combination with etoposide and genistein was checked to study their synergy on biofilm formation.

Confocal laser scan microscopy (CLSM) of biofilm: One-day biofilm was formed on a glass coverslip by incubating overnight grown *A. baumannii* culture in the absence and presence of etoposide, genistein (200 µM) or tobramycin (10× MIC) and their combination. Biofilm was treated with 1:100 dilution of 5µM Syto™ 9 and 10 µg/ml propidium iodide (PI) for 30 min in the dark for live and dead cell staining. Coverslip was washed with PBS (10 mM, pH 7.0) thrice and air-dried. Biofilm was observed under Nikon Eclipse – NI-E microscope. Signals were recorded at 517 nm for Syto™ 9 (green) after excitation at 488 nm and 636 nm for PI (red) after excitation at 493 nm. Images were further analyzed with ImageJ software (https://imagej.nih.gov/ij/download.html).

Surface motility: Low salt LB agar (0.3%) plates were prepared with and without 200µM of etoposide and genistein for motility assay. Overnight culture of *A. baumannii* was diluted 1: 100 with LB, incubated at 37°C and 180 rpm. Once the OD^{600nm} reach 1.0, the culture was washed and resuspended in 10 mM PBS. 1 µl of this cell suspension was inoculated in the center of low salt LB agar plates, incubated at 37°C for 36 h (Favre-Bonte et al., 2003) and zone of surface growth was observed.

Desiccation: Overnight grown *A. baumannii* culture was diluted 1: 100 with LB and incubated at 37°C and 180 rpm in the presence and absence of 200µM of etoposide and genistein till the cell density reached 10^8 CFUs/ml. From each sample, 10 µl of culture was spotted on sterilized nitrocellulose discs (0.2 µm, 6 mm diameter) pre-soaked in LB. The discs were incubated in a desiccator for 7 days (24°C and 48 - 52% RH) in the sterile Petri plates. Every alternate day, one disc was taken out form the desiccator, immersed in phosphate buffer (10 mM; pH 7.0), vortexed for 1 min and cell count for viable bacteria was determined by serial dilution method (Farrow et al., 2018).

2.14. Data analysis

Statistical analyses of the data were performed using GraphPad Prism 5.01 software (GraphPad Sofware Inc., CA, USA). Data were analyzed using a one-way ANOVA with Tukey's modification. *p*-value of ≤ 0.05 was considered statistically significant. The data represent mean ± SD (standard deviation) from three independent experiments (n = 3) and error bars represent the SD.

3. Results

3.1. Sequence analysis

The complete genome of *A. baumannii* ATCC 17978 (accession number CP000521) was obtained from NCBI. The 2082 bp gene encodes AbPPK1 at locus A1S_0990. Protein BLAST revealed that AbPPK1 consisting of 693 amino acids (UniProt ID: A3M3C8) belonged to the polyphosphate kinase superfamily of protein (Fig. S1). Non-redundant (nr) protein database search for maximum sequence conservation within *A. baumannii* (Taxonomy id: 470) revealed that AbPPK1 was conserved in different strains with > 98% query coverage (Fig. S2).

AbPPK1 is widely conserved among pathogenic and non-pathogenic bacteria and phylogenetic analysis revealed that AbPPK1 is close to PPK1 of *P. aeruginosa, Neisseria gonorrhoeae,* and *Burkholderia* spp., but relatively distant from *Helicobacter pylori* and *Campylobacter jejuni*. In due course of evolution, PPK1 from *E. coli, V. cholerae, Y. pestis, K. pneumoniae,* and *S. typhimurium* separated early (Fig. S3). AbPPK1 on pairwise sequence alignment showed 34% sequence identity with PPK1 from *E. coli* (UniProt ID: P0A7B1) and *Porphyromonas gingivalis* (UniProt ID: Q7MTR1). Amino acid sequence alignment showed that His-Ala-Lys (HAK) (443-445) tripeptide and Ala-Arg-Ile-Tyr (ARIY) (473-476) tetrapeptide lying in the pocket region were conserved in all the PPK1 sequences. The FLXI(F/Y)S (50-55) hexapeptide and Glu60 and Phe61 in the N-terminal region (α-2), Lys569, and Arg572 residues in the β-20 and EHXR region in β-22 were also conserved (Fig. S4).

3.2. Homology modeling of AbPPK1 and validation

5 models of AbPPK1 developed on the I-Tasser server, the best one with a TM score of 0.86 ± 0.07, signifying better precision for the topology of the model, was selected. The monomeric unit of AbPPK1 contained N-terminal, middle, and two C-terminal domains. Structure alignment of the AbPPK1 dimeric complex with *E. coli* PPK1 showed minimal deviation (RMSD = 0.251) (Fig. 1). VERIFY-3D predicted that 93% of residues of AbPPK1 had an average 3D-1D score of ≥ 0.2. Ramachandran plot for AbPPK1 showed 99.7% residues in the favorable and allowed region. Calculation of non-bonded interactions among different atom types achieved > 93% score (Table S2).

3.3. Binding site prediction and virtual screening

Binding site prediction using COATH and COFACTOR (I-Tasser server) identified Val33, Arg56, Ile53, Asn57, Glu60, Arg383, His443, Lys445, Asn472, Tyr 473, Ile475, Arg572, Arg595, Val597, His601, and Arg603 as the ATP binding residues in the pocket. The electrostatic potential analysis identified a positively charged pocket favorable for polyP binding. According to the UniprotKB database, His443 was predicted to be essential for the auto-phosphorylation of AbPPK1. Binding free energy (BFE) of ATP, the natural substrate of PPK, in the active site of AbPPK1 was -10.2 kcal mol^{-1}. After the virtual screening, seven compounds were selected based on interaction analysis (Table S3) for MD simulation.

3.4. MD simulation

Out of the seven compounds which showed good interaction with AbPPK1, only AbPPK1-etoposide, and AbPPK1-genistein complexes were found to be as stable as the AbPPK1-ATP complex. These complexes showed small increase in the root mean square deviation (RMSD) initially, which gradually stabilized as simulation proceeded towards completion. The average RMSD of AbPPK1-ATP, AbPPK1-genistein, and AbPPK1-etoposide were calculated in the range of 0.24, 0.3, and 0.32 nm, respectively (Fig. 2A). The superimposition of root mean square fluctuation (RMSF) values of ATP, genistein, and etoposide complexed with AbPPK1 revealed similar fluctuation patterns throughout the dynamic simulation and

RMSF ranged between 0.2 to 0.33 nm (Fig. 2B). While the radius of gyration of the structures was approximately 2.5 nm implying compactness and stability of the complexes. The simulation also showed improved binding affinities of ligands with active site residues of AbPPK1. Etoposide aligned with ATP in the cavity and despite being structurally more complex than ATP, it docked perfectly in the cavity (Fig. 3A, B). Post-simulation analysis of the AbPPK1-etoposide complex revealed that Leu249, Met623, and Arg630 of the AbPPK1 made additional interactions with etoposide and the orientation of the heterocyclic ring of etoposide was also changed (Fig. S5). In the AbPPK1-genistein complex, Arg54, Asp435, Asn546, Arg572, Phe599, and Ala620 of AbPPK1 showed additional interactions with genistein (Fig. S5) (Fig. 3A, C).

3.5. Cloning, hyperexpression and purification of AbPPK1 and its mutants

The insertion of the *ppk*1 gene into pET28a was confirmed by sequencing. *E. coli* BL 21 (DE3) cells transformed with pET28a-*ppk*1, expressed AbPPK1 when induced with 0.1 mM IPTG at 28˚C (Fig. S6A). The substitution of histidine with proline and alanine at 1329 and 1803 positions corresponding to H443P and H443A was confirmed by sequencing. Expressed proteins appeared as a single band of ~ 78 kDa on 10 % SDS PAGE after purification (Fig. S6B) and were used for biochemical analysis.

3.6. Biochemical characterization of AbPPK1 and its mutants

Phosphotransferase activity of AbPPK1, AbPPK1^{H443P}, and AbPPK1$^{H443P-H601A}$ was determined by quantitating ADP generated using ATP as the substrate. AbPPK1 was found to be most active at 30˚C and pH 7.0 (Fig S7A, B). AbPPK1 activity was better with Mg^{2+} than Mn^{2+} at 1mM (Fig. S7C). Single substitution mutant (AbPPK1^{H443P}) showed 46 % reduction in activity, whereas the double substitution mutant (AbPPK1$^{H443P-H601A}$) showed reduced activity by 83.6% (Fig. S7D).

3.7. CD of AbPPK1 and its mutants

The native structure of AbPPK1 displayed a strong negative shift in the 200 – 250 nm region (Fig 4A). Deviation in negative ellipticity started after 50˚ C, and further increase in temperature resulted in the denaturation of AbPPK1. T_m of AbPPK1 was calculated to be 52˚ C. In the presence of ATP (100 μM), T_m increased to 53˚ C, a similar increase was also observed in the presence of genistein, while etoposide shifted T_m to 56° C (Fig. 4B). Substitution mutants did not show any change in CD spectra (Fig. S8A, B).

3.8. Effect of etoposide and genistein on the activity of AbPPK1

Etoposide and genistein was shown to induce the AbPPK1 inhibition by ~ 80 % (IC_{50} = 27.53 μM) and ~ 72 % (IC_{50} = 24.38 μM) respectively (Fig. 5A). Michaelis-Menten kinetics revealed that AbPPK1 catalyzed reaction had K_m value 22.91 ± 1.84 μM and V_{max} 10.73 ± 0.23 μM/min/mg. Etoposide and genistein significantly increased the K_m to 27.41 ± 2.68 μM and 25.95 ± 2.4 μM, respectively (Fig. 5B, C). Lineweaver-Burk plot suggested a competitive mode of inhibition of AbPPK1 by the drugs (Fig. 5D).

3.9. Effect of etoposide and genistein on the expression of ppk1, ppx, phoU and rpoS

Genistein increased the expression of *ppk*1 by approximately 1.4 fold, while an insignificant increase in the expression was observed in the presence of etoposide. However significant increase ($p < 0.005$) in the expression of *ppx* was observed by etoposide (~2 fold) and genistein (1.7 fold) (Fig. 6A). Concomitantly there was decrease in the intracellular polyP content as compared to untreated *A. baumannii* ATCC 17978 (Fig. 6B). Expression of *pho*U was upregulated by approximately 1.8 and 1.2 fold with genistein and etoposide respectively. *Rpo*S, the stress response gene was down-regulated by 2.2 fold by genistein.

3.10. Effects of etoposide and genistein on virulence factors

3.10.1. Biofilm formation

Biofilm formation contributes largely to chronic hospital-acquired infections. Therefore, we studied the effects of etoposide and genistein on biofilm formation. No inhibition of *A. baumannii* ATCC 17978 growth was observed till 200μM, however, at higher concentration (500 μM), growth inhibition was seen in the case of etoposide (Fig. S9). Hence these drugs were used at ≤ 200 μM concentration to check their effect on biofilm and other virulence traits. Etoposide and genistein (200 μM) reduced the biofilm index (BI) by 2.2 and 1.5 fold, respectively (Fig. 7A). Significant inhibition of biofilm formation (1.8 - 2.4 fold) was also observed in three clinical isolates (Ci-1, Ci-2, and Ci-4) of *A. baumannii* (Fig. 7B). Although biofilm formation was also reduced in clinical isolate 3 (Ci-3) but the inhibition was non-significant.

CLSM showed that *A. baumannii* formed thick, mat-like biofilm on glass coverslip, which was reduced to fragmented aggregates of cells in the presence of etoposide and genistein (Fig. 7C). The MIC of tested antibiotics (Table S4) was used to check their effect, and among these, tobramycin was most effective in reducing biofilm formation by approximately 2 fold at 10× MIC (MIC = 2.0 µg/ml) (Fig. S10). However, etoposide in combination with tobramycin, did not significantly lower the BI (data not shown), but the combination of genistein with tobramycin reduced the biofilm significantly, and only a few dispersed individual cells were observed. Quantitative analysis of CLSM images revealed that biofilms treated with either etoposide, genistein, or their combination with tobramycin had a similar proportion of live and dead cells, which indicated that reduction in biofilm formation was not because of the enhanced killing of cells (Fig. S11). The coefficient of variation of the averaged intensities was calculated from randomly selected areas of green and red fluorescence, which fall in the range of 0.3 - 1.2 and 0.21 - 0.8, respectively.

3.10.2. Acyl homoserine lactone (AHL) production

Etoposide significantly reduced AHL production by 3.8 fold in *A. baumannii* ATCC 17978 and also in the clinical isolates; Ci-1, Ci-2, Ci-3, and Ci-4 (2 - 4 fold), whereas genistein did not show significant reduction in AHL production (Fig. 8A, B).

3.10.3. Surface motility

Treatment with etoposide and genistein significantly reduced the surface motility of *A. baumannii* strain ATCC 17978 and the clinical isolates; Ci-1, Ci-2, Ci-3, and Ci-4 (Fig. 8C).

3.10.4. Desiccation

Loss in viability was observed from 7.9 Log_{10} CFU/ml to approximately 2 Log_{10} CFU/ml on the 7th day of desiccation. Etoposide and genistein effectively enhanced (p < 0.05) the sensitivity to desiccation and reduced the viable cell count to ~ 0.21 and 0.7 Log_{10} CFU/ml, respectively (Fig. 8D). Spot assay also confirmed the enhanced sensitivity of *A. baumannii* ATCC 17978 to desiccation in their presence (Fig. 8E).

4. Discussion

PPK1 is involved in the pathogenesis of *E. coli, P. aeruginosa, F. tularensis, M. tuberculosis,* and other pathogens (Ogawa et al., 2000; Rohlfing et al., 2017), but only a few studies are available on PPK1 of *A. baumannii* (Itoh and Shiba, 2004; Shiba et al., 2005). In the present study, we found that the primary sequence of AbPPK1 had high degree of similarity with PPK1 of most pathogenic organisms and also shared close evolutionary relationships with them. Here, AbPPK1 homology model was used for virtual screening of inhibitors from a library composed of natural compounds and FDA approved drugs. Etoposide, a plant alkaloid, approved for cancer chemotherapy (Soo et al., 2016), showed stable binding in the catalytic core of AbPPK1 during the molecular dynamic simulation. Since etoposide is already an approved bioavailable drug, it can be sought from the point of repurposing. Genistein, a plant flavonoid that is an inhibitor of topoisomerase II and protein tyrosine kinase (Mizushina et al., 2013; Zhou et al., 2009), also showed stable binding with AbPPK1. Genistein has been reported to exhibit antibacterial effect (Choi et al., 2018; Hong et al., 2006) and at 50 μg/ml significantly inhibited biofilm formation in *A. baumannii* due to the blocking of a response regulator, BfmR of a two-component signal system (Raorane et al., 2019).

Substitution mutant of AbPPK1 in which histidine residues at positions 443 and 601 were replaced with proline and alanine, respectively, showed poor kinase activity as compared to the wild type. Substitution of these conserved histidine residues had previously been shown to disrupt PPK1 activity in *E. coli* (Kumble et al., 1996; Zhu et al., 2005) and *P. aeruginosa* (Ishige et al., 1998). The initial step of polyP synthesis is the auto-phosphorylation of PPK1 (Ahn and Kornberg, 1990), and homologous histidine residues in *E. coli* were found to be critical for phosphotransfer of terminal phosphate from ATP (Zhu et al., 2003, 2005), which probably is true for AbPPK1 also.

Usually, compounds showing high affinity for kinases are considered as promising scaffolds for therapeutic development (Vivoli et al., 2015). Thermal stability assays, which are used for screening and validation of protein-ligand interactions in the solution (Hussain and Siligardi, 2016; Huynh and Partch, 2015), showed that there was no change in the overall secondary structure of AbPPK1 in the presence of etoposide and genistein. These drugs formed the thermally stable complexes with AbPPK1 and were effective in inhibiting its activity in competitive manner at micromolar concentration.

Apart from PPK1, the role of exopolyphosphatase (PPX) is also crucial in polyP homeostasis inside the cell (Gallarato et al., 2014; Thayil et al., 2011). In *E. coli*, overexpression of *ppk* and *ppx* led to the reduction of intracellular polyP and increased tolerance to heavy metals (Keasling et al., 1998). Over-expression of *ppx* and depletion of polyP resulted in the loss of viability of *E. coli* (Gray et al., 2014). In *P. aeruginosa*, PPX has been suggested to be important for motility and production of virulence factors such as biofilm, rhamnolipids, and quorum sensing molecules (Gallarato et al., 2014). In *A. baumannii,* etoposide and genistein led to reduced intracellular polyP with significant reduction in motility, biofilm formation, and desiccation tolerance, which could be attributed by inhibition of AbPPK1 activity and increased expression of PPX. We studied the expression of phoU, a component of pst operon that regulates inorganic phosphate transport through pstSCAB complex (Rice et al., 2009). PhoU act as a repressor of Pho regulon that is involved in the control of virulence in *P. aeruginosa* (Munévar et al., 2017). It also modulates the expression of PPK and PPX and *pho*U mutants showed large accumulation of polyP (de Almeida et al., 2015; Munévar et al., 2017). In the present study, both genistein and etoposide upregulated the expression of *pho*U which might have also contributed to decrease in polyP level in *A. baumannii* ATCC 17978.

PolyP has already been known to be involved in many of physiological processes, such as motility, biofilm development, quorum sensing (QS), which are related to bacterial pathogenesis (Rashid et al., 2000; Rashid and Kornberg, 2000; Seufferheld et al., 2008; Shi et al., 2004). Etoposide reduced the production of QS signal molecule, N-3-hydroxy-dodecanoyl-homoserine lactone (N-3-OH-C_{12}-HSL), which also affect biofilm production and motility in *A. baumannii* (Chow et al., 2014; Stacy et al., 2012). Since the QS signal is essential for exopolysaccharide (EPS) production in bacteria (Decho, 2000), which protects them against desiccation (Costa et al., 2018), the reduced production of AHL in the presence of etoposide may be responsible for decrease in survival of *A. baumannii* under desiccation.

Although both inhibitors increased the sensitivity of *A. baumannii* to desiccation at sub-inhibitory concentration, their mode of action appears to be different. Genistein downregulated the expression of *rpo*S that encodes RNA polymerase σ-38, that controls the transcription of a plethora of genes necessary for the entry of cells into the stationary phase or to cope up with stress conditions (Battesti et al., 2015; Landini et al., 2014; McMeechan et al., 2007). Regulation of *rpo*S by PPK1 seems to be species-specific, as its expression differed in the case of *ppk*1 deletion in *E. coli* (Shiba et al., 1997), *Salmonella typhimurium* (Cheng and Sun, 2009), and *B. pseudomallei* (Srisanga et al., 2019; Wongtrakoongate et al., 2012), while there was no change in its expression in *P. aeruginosa* (Bertani et al., 2003).

Genistein, in combination with tobramycin synergistically reduced biofilm formation by *A. baumannii*. In *S. aureus,* genistein has been reported to exert efflux pump inhibitor activity against the NorA pump resulting in reduction in the MIC of norfloxacin and berberine (Morel et al., 2003; Stermitz et al., 2002). Since live dead staining of *A. baumannii* biofilm did not show enhanced killing in the presence of subinhibitory concentration of genistein, the reduction in biofilm formation could be mediated through inhibition of AbPPK1.

5. Conclusion

The present study focuses on the characterization of AbPPK1 for the repurposing of drugs targeting its activity. Etoposide and genistein inhibited AbPPK1 activity, upregulated *ppx* expression, and concomitantly reduced the total cellular polyP in *A. baumannii*. There was significant reduction in biofilm formation, surface motility and reduced survival under desiccation by these drugs. The use of sub-inhibitory concentration of etoposide and genistein to attenuate the virulence factors would reduce the risk of developing drug resistance and can be explored as new treatment option in combination with antibiotics.

Declaration of interest

The authors declare no conflict of interest, financial or otherwise.

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Figure Captions

Fig. 1: Model structure of AbPPK1: (A) N-terminal domain (green), the 'head' or middle domain (red) and two C-terminal domains: C1 (cyan) and C2 (purple-blue); (B) structure alignment of AbPPK1 with *E. coli* PPK1 (1XDP) (magenta), RMSD = 0.251 Å.

Fig. 2: (A) All-atom root mean square deviation (RMSD) and (B) root mean square fluctuations (RMSF) after performing a 20ns molecular dynamic simulation of *Ab*PPK1-ATP and *Ab*PPK1-etoposide complexes.

Fig. 3: (A) Electrostatic map of AbPPK1 monomeric unit showing ATP binding cavity (inset). Docking of etoposide (B) and genistein (C) in the ATP binding cavity. Electrostatic potential: – 2 (red), +2 (blue), and 0 (white).

Fig. 4: (A) CD spectra of AbPPK1 recorded in the temperature range of 20 - 80 ̊C. The CD data were expressed as mean residue ellipticity with the unit of 10^3 deg cm^2 $dmol^{-1}$; (B) Thermal stability of AbPPK1 in the presence and absence of ATP, etoposide and genistein. The CD spectrum was obtained from the data sets of three replicate experiments for each treatment. The mean residue ellipticity (θ) was fit using GraphPad Prism 5.01 software.

Fig. 5: (A) Inhibition profile of AbPPK1 with drugs. To calculate the IC_{50} values for each drug, a dose-response curves fitted by non-linear regression was generated analysis. Assay was performed in triplicate (n = 3) and results are shown as the mean of the normalized values ± SD. Michaelis - Menten kinetics of recombinant AbPPK1 in the absence and presence of different concentration of (B) genistein and (C) etoposide; ATP was used as the substrate; Each data point represents mean ± SD. Error bars represent SD; (*n* = 3). (D) Lineweaver-Burk plots for inhibition of AbPPK1 activity in the presence of etoposide and genistein.

Fig. 6: (A) Relative expression of *ppk1*, RNA polymerase sigma factor (*rpo*S), exopolyphosphatase (*ppx*) and *pho*U component of phosphate regulon on treatment with etoposide and genistein in the late log phase cells of *A. baumannii* ATCC 17978. Untreated cells were taken as control with basal level expression indicated as 1. The data is representative of three independent experiments in which each sample was assayed in duplicate. Statistical analysis was done using Tukey's test with *p*-value ≤ 0.05. Bars represent the mean ± SD. **p*≤ 0.05; ***p* ≤ 0.01; ****p* ≤ 0.001; ns as non-significant. **(**B) PolyP levels in *A. baumannii* at different time points of growth. The fluorescence intensity of DAPI-polyP complex is presented as arbitrary units (AU) (excitation at 415 nm and emission at 550 nm). The data is representative of mean ± SD; (n= 3). Error bars represent the standard deviation.

Fig. 7: (A) Effects of inhibitors at different concentration (50 µM, 100 µM, and 200 µM) on biofilm index ($OD_{570/600}$). The data is representative of mean ± SD; n = 3. The data was tested with one way ANOVA and *p*-value ≤ 0.05 was considered as significant. Error bars represent SD. (B) Effects of etoposide and genistein (200 µM) on biofilm index ($OD_{670/600}$) of *A. baumannii* ATCC 17978 and clinical isolates; Ci-1, Ci-2, Ci-3, and Ci-4. The data is

representative of mean ± SD; n = 3. Analysis was done with one way ANOVA and *p*-value ≤ 0.05 was considered as significant. Error bars represent SD. (C) Confocal laser scanning microscopy images of biofilm formed by *A. baumannii* ATCC 17978 on glass coverslip in the absence (i) and presence of etoposide (200 µM) (ii), genistein (200 µM) (iii), tobramycin (10× MIC) (iv), and genistein + tobramycin (v) by live (Syto9) and dead (PI) staining. Fluorescence intensities quantified using NIS element viewer (Nikon) are shown below the respective images from the area indicated by red arrows. Lower panel images are iso-surface rendered 3D surface plots of the same data set quantified using ImageJ software (v1.44p).

Fig. 8: (A) β-galactosidase activity by AHL produced by *A. baumannii* grown in the absence and presence of etoposide and genistein. (B) Production of AHL by etoposide treated and untreated *A. baumannii* ATCC 17978 and clinical strains *$p \leq 0.05$, *** $p \leq 0.0005$. (C) Surface motility of *A. baumannii* on LB agar (0.3%) plates in the absence and presence of etoposide and gensitein. Desiccation survival assay: *A. baumannii* ATCC 17978 in the presence and absence of etoposide and genistein was tested for survival after drying. (D) Cells from the culture of *A. baumannii* were harvested after 0, 1, 3, 5 and 7 days, washed with water, dried, and survival was assessed. The data presented the mean ± SD from three independent experiments and error bars represent the standard deviation.*$p = 0.0083$. (D) Serially diluted samples were spotted onto the surface of an LB plate and incubated at 37˚C.

Fig. 1

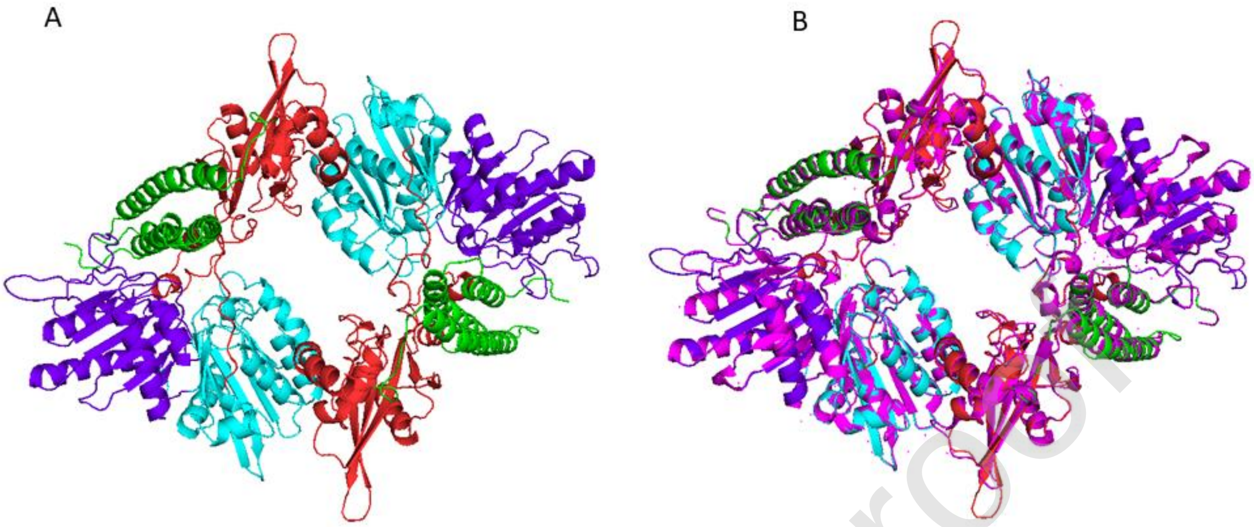


Fig. 2

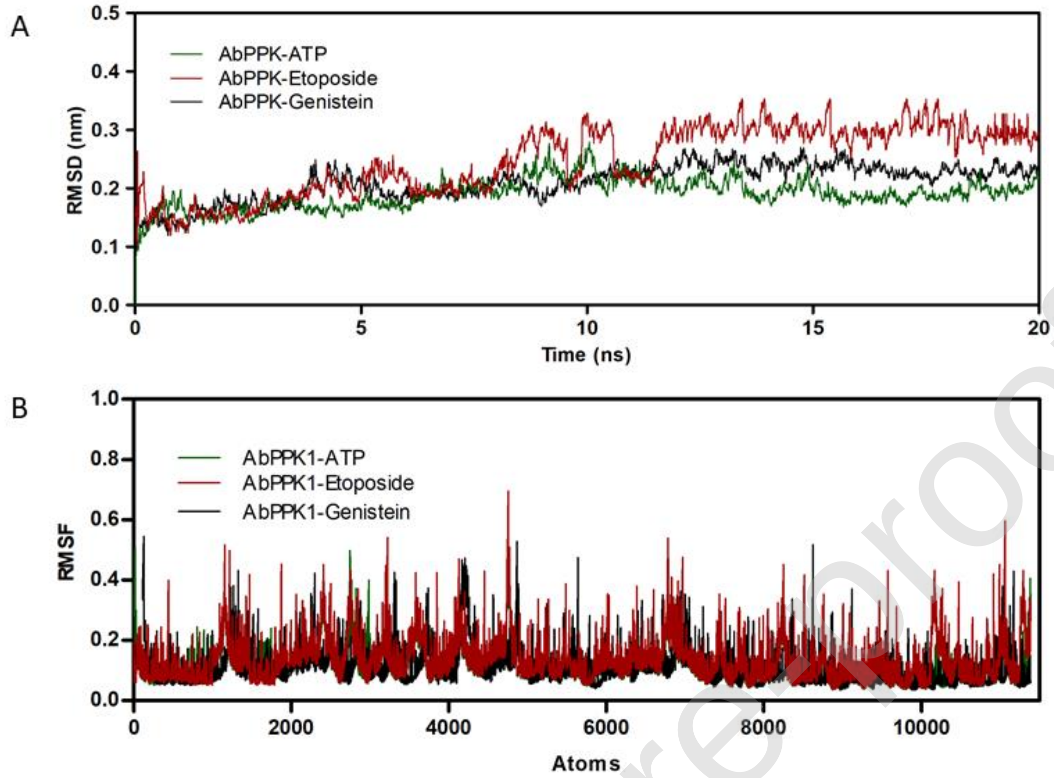

A
0.5
0.4
0.3
0.2
0.1
0.0
RMSD (nm)
AbPPK-ATP
AbPPK-Etoposide
AbPPK-Genistein
0
5
10
15
20
Time (ns)
B
1.0
0.8
0.6
0.4
0.2
0.0
RMSF
AbPPK1-ATP
AbPPK1-Etoposide
AbPPK1-Genistein
0
2000
4000
6000
8000
10000
Atoms

Fig. 3

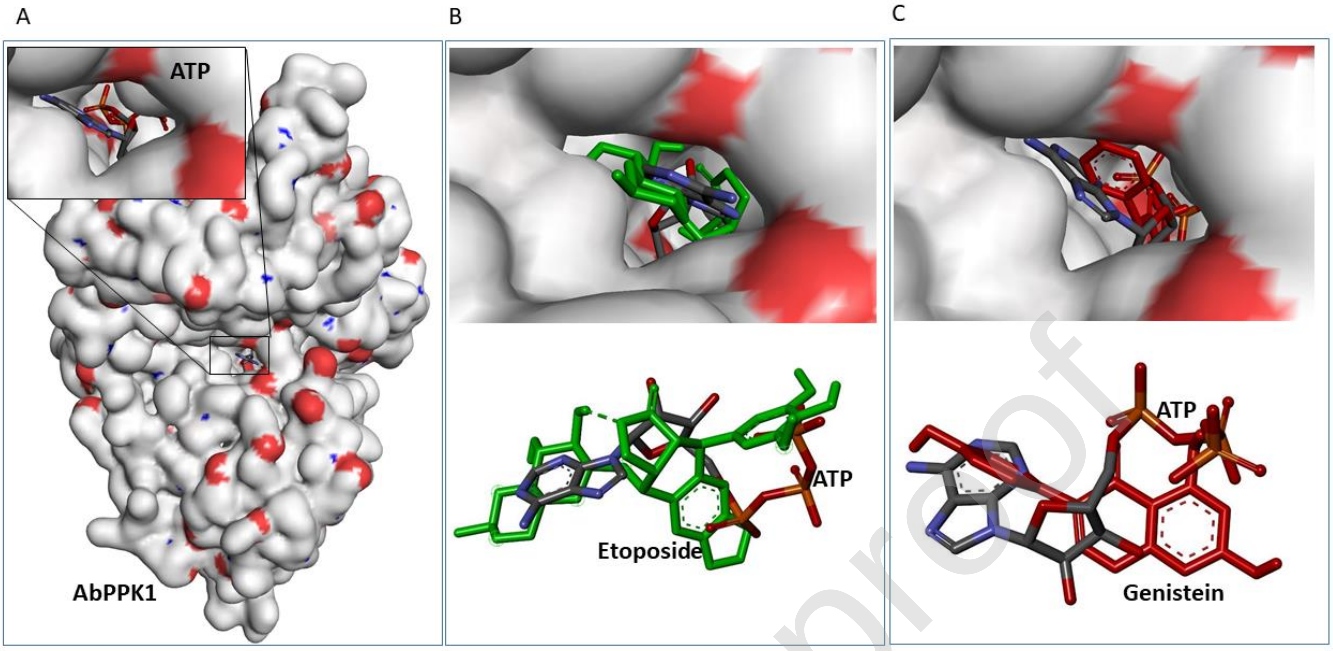


Fig. 4

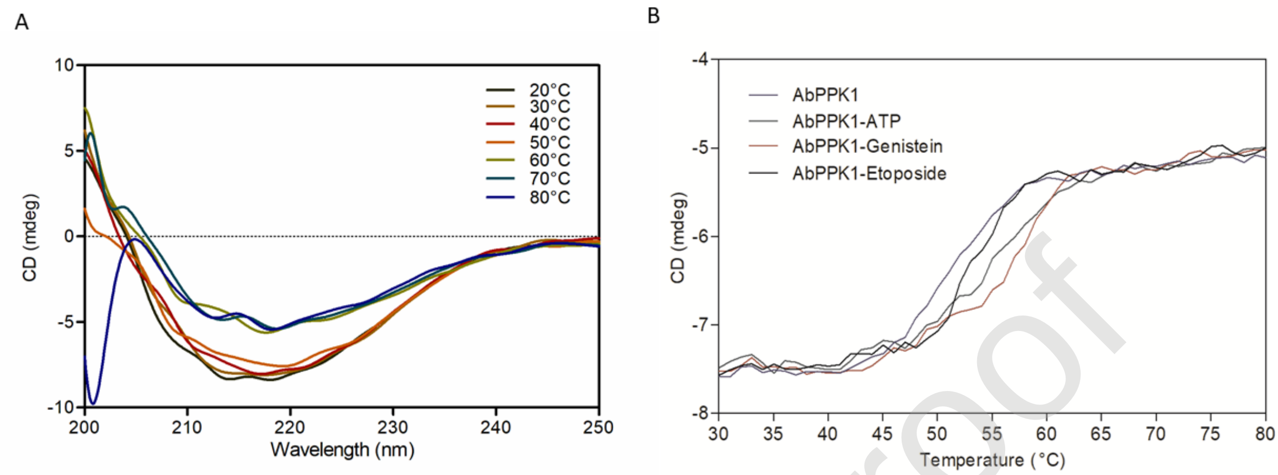

A
10
5
0
-5
-10
200
210
220
230
240
250
Wavelength (nm)
CD (mdeg)
20°C
30°C
40°C
50°C
60°C
70°C
80°C
B
-4
-5
-6
-7
-8
30
35
40
45
50
55
60
65
70
75
80
Temperature (°C)
CD (mdeg)
AbPPK1
AbPPK1-ATP
AbPPK1-Genistein
AbPPK1-Etoposide

Fig. 5

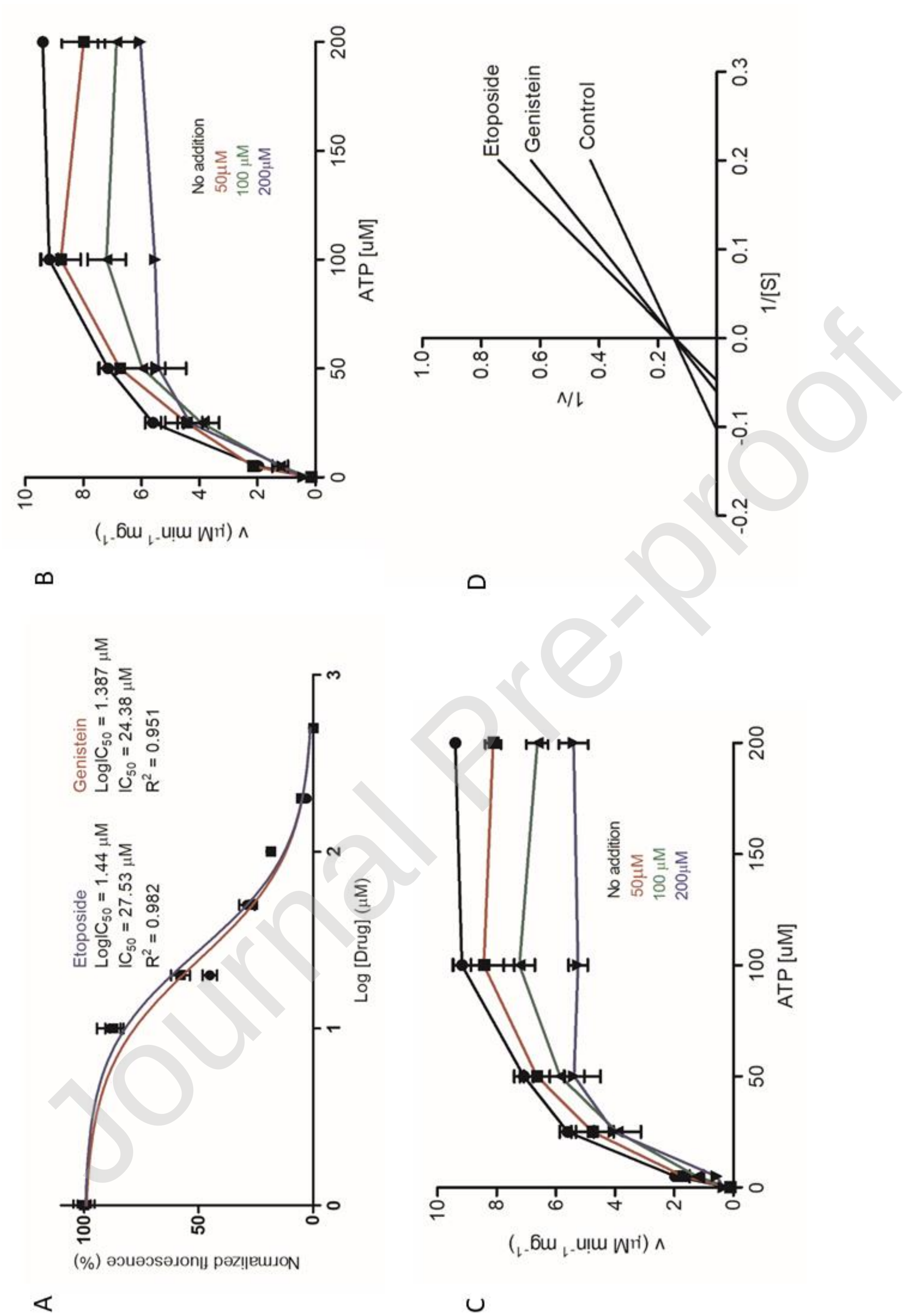

A
Etoposide
LogIC50 = 1.44 µM
IC50 = 27.53 µM
R2 = 0.982
Genistein
LogIC50 = 1.387 µM
IC50 = 24.38 µM
R2 = 0.951
Normalized fluorescence (%)
Log [Drug] (µM)
B
v (µM min-1 mg-1)
ATP [uM]
No addition
50µM
100 µM
200µM
C
v (µM min-1 mg-1)
ATP [uM]
No addition
50µM
100 µM
200µM
D
1/v
1/[S]
Etoposide
Genistein
Control

Fig. 6

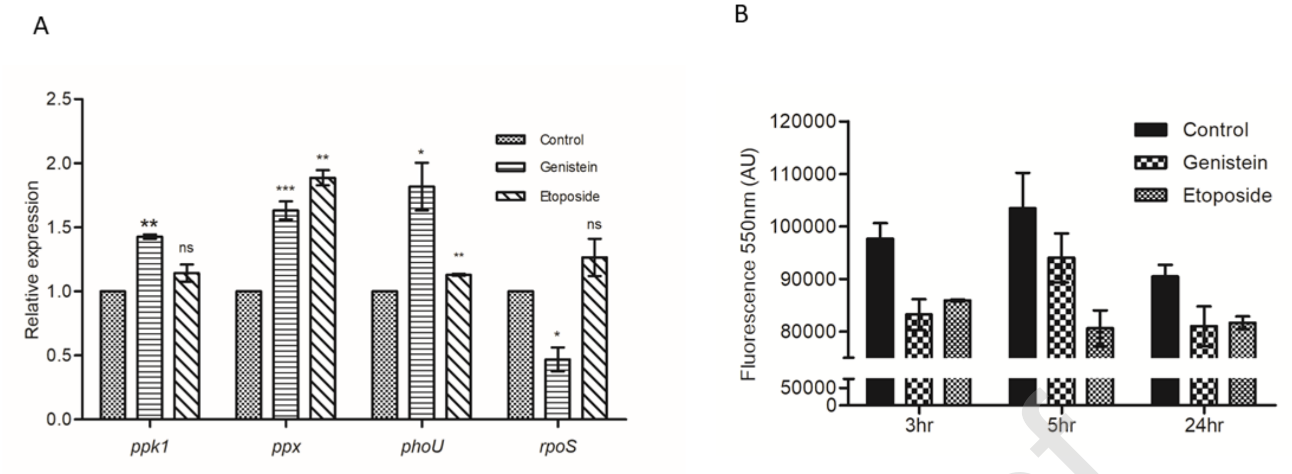

A
B
Relative expression
2.5
2.0
1.5
1.0
0.5
0.0
Control
Genistein
Etoposide
**
ns
***
**
*
**
*
ns
ppk1
ppx
phoU
rpoS
Fluorescence 550nm (AU)
120000
110000
100000
90000
80000
50000
0
3hr
5hr
24hr

Fig. 7

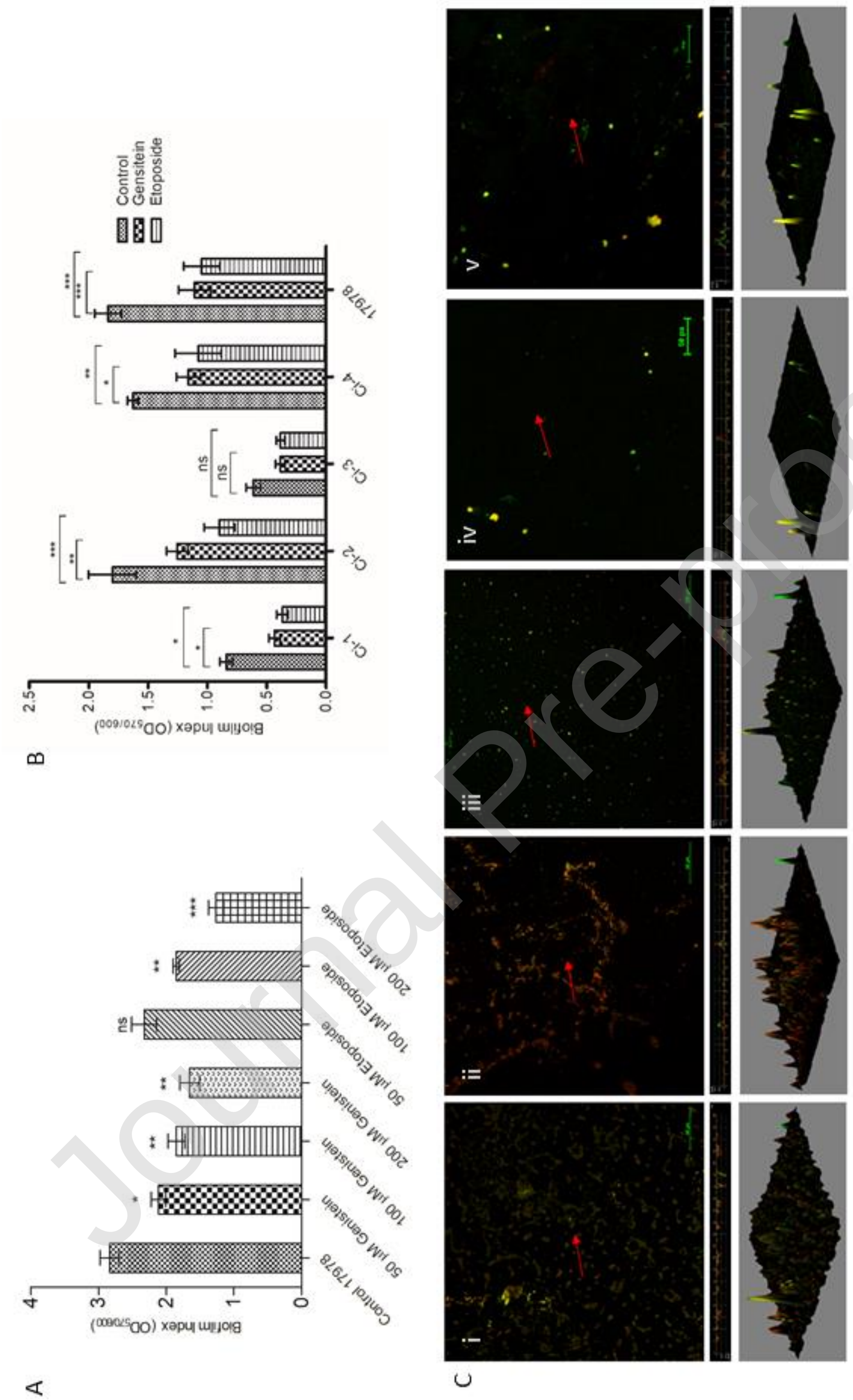
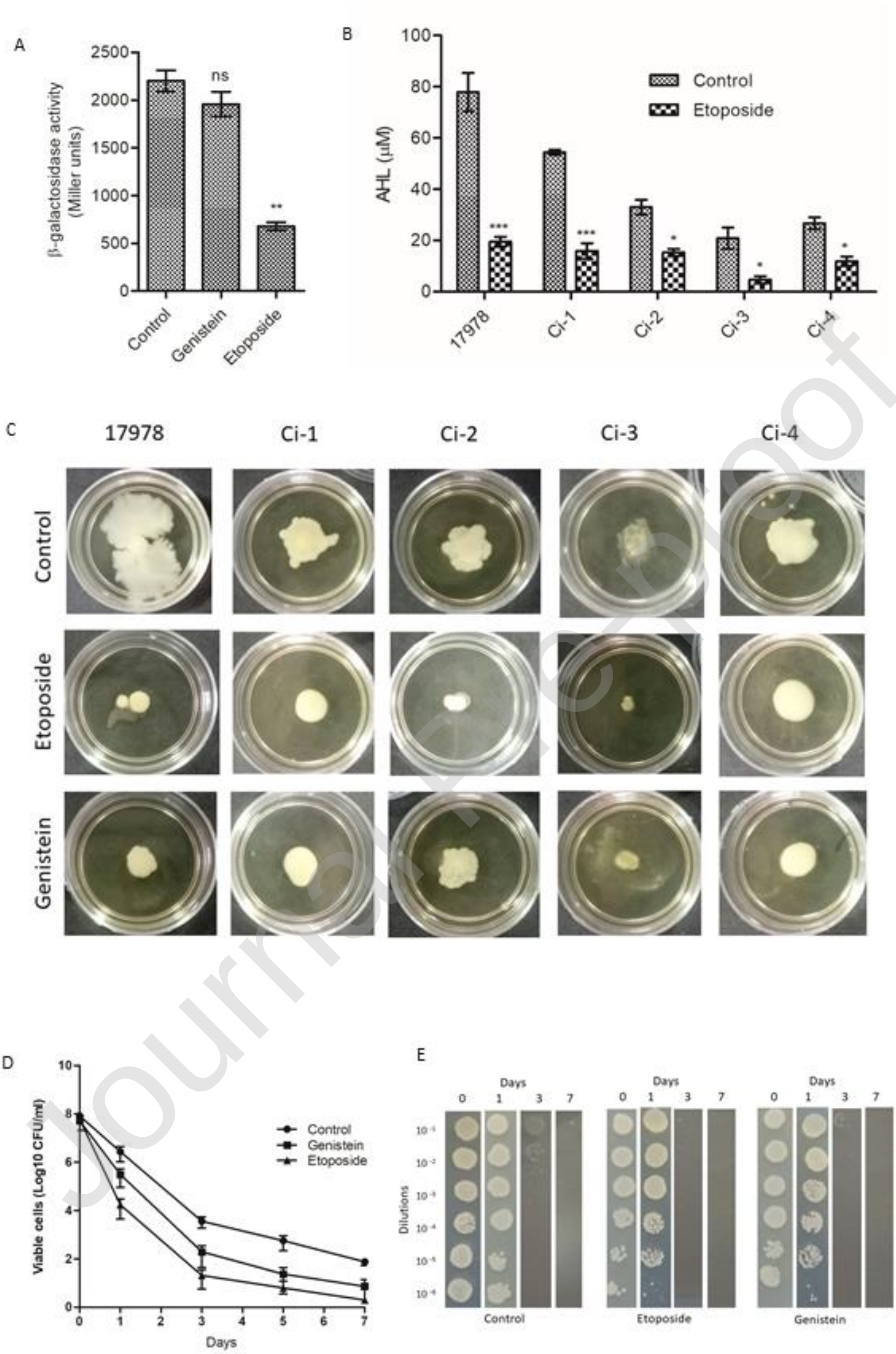

A
Biofilm Index (OD570/600)
Control 17978
50 µM Genistein
100 µM Genistein
200 µM Genistein
50 µM Etoposide
100 µM Etoposide
200 µM Etoposide
B
Biofilm Index (OD570/600)
Control
Gensitein
Etoposide
Cl-1
Cl-2
Cl-3
Cl-4
17978
C
i
ii
iii
iv
v

Fig. 8


A
β-galactosidase activity (Miller units)
2500
2000
1500
1000
500
0
ns
**
Control
Genistein
Etoposide
B
AHL (μM)
100
80
60
40
20
0
Control
Etoposide
***
***
*
*
*
17978
Ci-1
Ci-2
Ci-3
Ci-4
C
17978
Ci-1
Ci-2
Ci-3
Ci-4
Control
Etoposide
Genistein
D
Viable cells (Log10 CFU/ml)
10
8
6
4
2
0
Control
Genistein
Etoposide
0
1
2
3
4
5
6
7
Days
E
Days
0
1
3
7
Dilutions
Control
Etoposide
Genistein